Bike Sharing System Analysis and Visualization

1. Project Overview

- The Bike Sharing System Analysis project focuses on analyzing bike availability, station distribution, and operational efficiency across multiple cities. The dataset is processed and visualized using Power BI to generate insights that support better resource allocation and system expansion.

  This project helps understand:

  - Bike availability patterns
  - Station distribution across cities
  - Infrastructure gaps (banking, bonus facilities)
  - Growth trends over time

2. Project Objectives

- To analyze bike availability across cities and stations
- To identify high-demand and low-availability stations
- To study infrastructure availability (banking, bonus, open/closed stations)
- To understand geographic distribution of bike stations
- To analyze growth trends in bike sharing systems
- To provide recommendations for system improvement

3. Data Source

- The dataset contains bike sharing information collected from multiple cities
- Includes station-level operational and geographic data

Attribute Name	Description	Data Type
City_Name	Name of the city	Text
Station_Name	Name of the bike station	Text
Station_Number	Unique station identifier	Number
Address	Location details of station	Text
Latitude	Geographic coordinate	Decimal
Longitude	Geographic coordinate	Decimal
Bike_Availability	Number of available bikes	Number
Stands	Total parking stands	Number
Banking	Availability of banking facility	Boolean/Text
Bonus	Bonus facility availability	Boolean/Text
Status	Station status (Open/Closed)	Text
Timestamp	Date & time of record	DateTime

4. Problem Statement

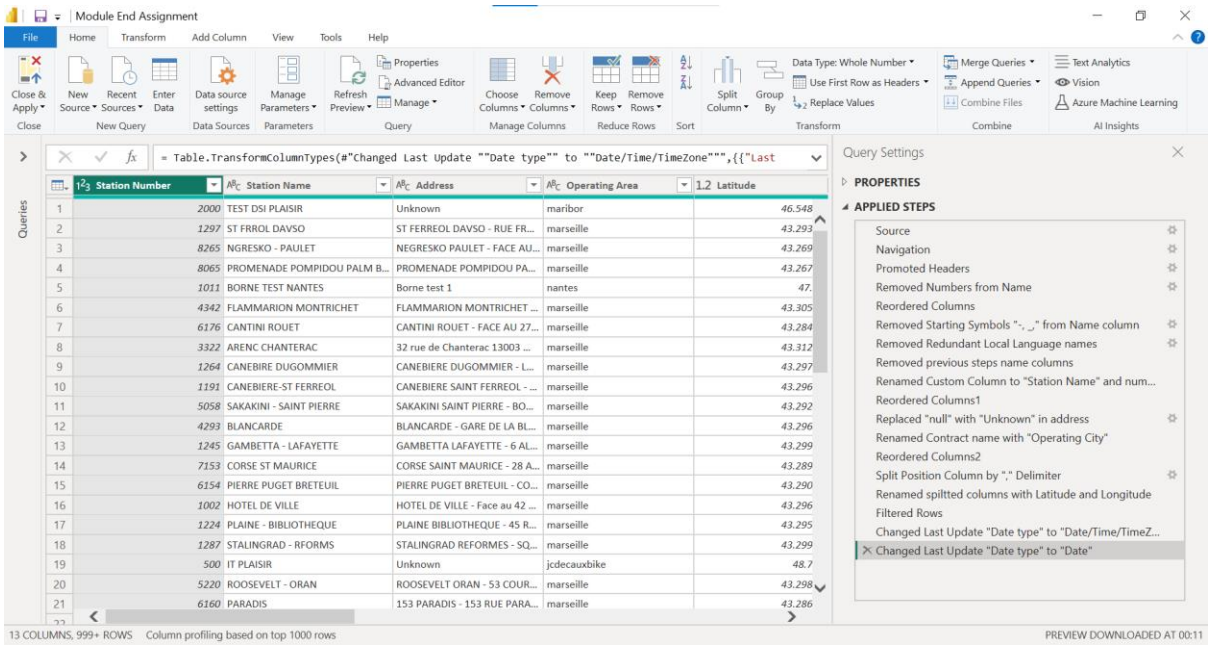
- To analyze bike availability patterns across different cities and stations
- To identify imbalance in bike distribution across stations
- To detect stations with high demand but low availability
- To evaluate infrastructure gaps such as lack of banking or bonus facilities
- To study station status and identify inactive/closed stations
- To analyze growth trends in bike-sharing systems
- To support decision-making for system expansion and optimization

5. Data Import

- The bike sharing dataset was imported into **Power BI** for analysis.
- Columns were examined to understand the available information such as city name, station name, bike availability, and timestamps.

6. Data Pre-Processing (Power BI / Power Query)

Data Cleaning & Transformation



- It was observed that the Station Number column already contained the numeric identifier, and the same number appeared inside the Station Name column. To avoid redundancy, the numbers were removed from the station name field.
- Special characters such as "-" were removed from the station name column to improve readability.
- Some station names contained local language text along with English names. The unnecessary local language text was removed to standardize the column values.
- In the Address column, some values were marked as null. These were replaced or standardized to maintain consistency in the dataset.
- The Position column contained combined values of latitude and longitude. This column was split using a delimiter to create two separate columns Latitude and Longitude:
- Data types were corrected for accuracy:

- Unnecessary or irrelevant rows were filtered out to improve data quality.

7. Data Modelling

Creating Dimension and Fact Tables:

- After completing data cleaning in **Power BI**, the cleaned dataset was exported and used in **Excel** to structure the data into **Fact and Dimension tables**. This step was performed to build a proper **Star Schema data model** for efficient analysis.
- **Schema Type: Star Schema**
- The dataset was structured into fact and dimension tables to improve performance and analytical capability.
- **Dim_Date Table:** Contains date-related attributes derived using below excel Date formula.

```
IF(Date(2022,1,1)+Row(A2)-2<=Date(2025,31,12), Date(2022,1,1)+Row(A2)-2,"")
```

Date	Year	Month	Month Nam	Day
01-01-2022	2022	1	Jan	1
02-01-2022	2022	1	Jan	2
03-01-2022	2022	1	Jan	3
04-01-2022	2022	1	Jan	4
05-01-2022	2022	1	Jan	5
06-01-2022	2022	1	Jan	6
07-01-2022	2022	1	Jan	7
08-01-2022	2022	1	Jan	8
09-01-2022	2022	1	Jan	9
10-01-2022	2022	1	Jan	10
11-01-2022	2022	1	Jan	11
12-01-2022	2022	1	Jan	12
13-01-2022	2022	1	Jan	13
14-01-2022	2022	1	Jan	14
15-01-2022	2022	1	Jan	15
16-01-2022	2022	1	Jan	16
17-01-2022	2022	1	Jan	17
18-01-2022	2022	1	Jan	18
19-01-2022	2022	1	Jan	19

- **Dim_Station:** Contains station-level information like below screenshot

Address	Station Name	Latitude	Longitude	Station-
AVENIDA DE LLANES - Aprox. Avda de Miraflores	AVENIDA DE LLANES	37.40513	-5.97483	S01
PARLAMENTO - Aprox. C/ Don Fadrique	PARLAMENTO	37.40333	-5.98904	S02
RONDA TAMARGUILLO - Aprox. Alfonso Laso de la Vega	RONDA TAMARGUILLO	37.36705	-5.97110	S03
CALLE TESALÓNICA - Calle Mazzantini	CALLE TESALNICA	37.39609	-5.95876	S04
AVENIDA ALCALDE MANUEL DEL VALLE - Aprox. C/ Alejandro Sawa	AVENIDA ALCALDE MANUEL DEL VALLE	37.40509	-5.96886	S05
CALLE ENRAMADILLA - Aprox. C/ Balbino Marrón	CALLE ENRAMADILLA	37.37857	-5.97837	S06
GLORIETA REPUBLICA DOMINICANA - Aprox. C/ López de Gomara	GLORIETA REPUBLICA DOMINICANA	37.37574	-6.00568	S07
CALLE ESTRELLA CANOPUS - Aprox. C/ estrella Betelgeuse	CALLE ESTRELLA CANOPUS	37.42457	-5.96999	S08
CALLE URQUIZA - Aprox. C/ Venecia	CALLE URQUIZA	37.39454	-5.97990	S09
RONDA TAMARGUILLO - Aprox. Marqués de Pickman	RONDA TAMARGUILLO	37.38069	-5.96050	S10
CALLE DOCTOR JAIME MARCOS - Aprox. Avda. Doc. Fedriani	CALLE DOCTOR JAIME MARCOS	37.40966	-5.98425	S11
CALLE PAEZ DE RIVERA - Aprox. Paseo de la Palmera	CALLE PAEZ DE RIVERA	37.36362	-5.98479	S12
AVENIDA DE CORIA - Aprox. C/ Coruña	AVENIDA DE CORIA	37.38111	-6.01201	S13
CALLE SAN JUAN BOSCO - Aprox. C/ Arroyo	CALLE SAN JUAN BOSCO	37.39675	-5.97863	S14
AVENIDA EDUARDO DATO - Detrás Estadio Sánchez Pizjuan	AVENIDA EDUARDO DATO	37.38265	-5.97113	S15
CALLE PINO MONTANO - Aprox. C/ Estafeta	CALLE PINO MONTANO	37.41612	-5.96756	S16
CALLE DOCTOR LAFFON - Aprox. C/ Efeso	CALLE DOCTOR LAFFON	37.39540	-5.96404	S17
AVENIDA KANSAS CITY - Aprox. Pza el Tato	AVENIDA KANSAS CITY	37.39997	-5.96185	S18
CALLE AMADOR DE LOS RIOS - Aprox. C/ Lope de Vega	CALLE AMADOR DE LOS RIOS	37.39071	-5.98277	S19
AVDA ESPERANZA DE TRIANA - Aprox. Plaza Amadeo Janone	AVDA ESPERANZA DE TRIANA	37.38015	-6.00438	S20

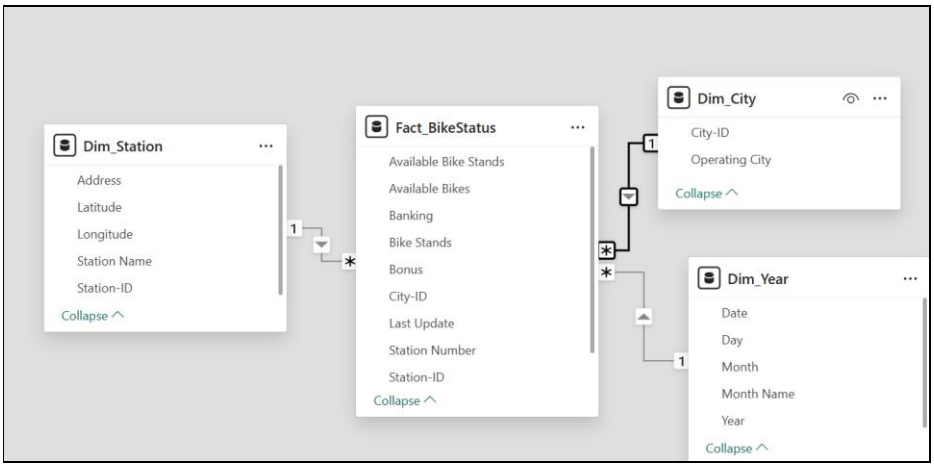
- **Dim_City:** Contains city-related information shown as below

Operating City	City-ID
Seville	C01
Valence	C02
Luxembourg	C03
Lyon	C04
Dublin	C05
Bruxelles-Capitale	C06
Ljubljana	C07
Toulouse	C08
Besancon	C09
Toyama	C10
Maribor	C11
Amiens	C12
Nancy	C13
Nantes	C14
Lund	C15
Namur	C16
Vilnius	C17
Jcdecauxbike	C18
Lillestrom	C19

- **Fact_BikeStatus:** The Fact_BikeStatus table contains the main operational data related to bike availability and station status.
    - This table stores the measurable and transactional data used for analysis.

Station Number	Banking	Bonus	Status	Bike Stands	Available Bike Stands	Available Bikes	Last Update	City-ID	Station-ID
22	FALSE	FALSE	OPEN	20	18	1	12-Nov-25	C01	S01
19	FALSE	FALSE	OPEN	20	18	2	12-Nov-25	C01	S02
172	FALSE	FALSE	OPEN	20	7	13	12-Nov-25	C01	S03
203	FALSE	FALSE	OPEN	20	2	9	12-Nov-25	C01	S04
185	FALSE	FALSE	OPEN	20	19	1	12-Nov-25	C01	S05
122	FALSE	FALSE	OPEN	20	1	18	12-Nov-25	C01	S06
130	FALSE	FALSE	OPEN	20	0	20	12-Nov-25	C01	S07
196	FALSE	FALSE	OPEN	20	12	8	12-Nov-25	C01	S08
151	FALSE	FALSE	OPEN	20	20	0	12-Nov-25	C01	S09
174	FALSE	FALSE	OPEN	20	13	7	12-Nov-25	C01	S10
180	FALSE	FALSE	OPEN	20	17	1	12-Nov-25	C01	S11
132	FALSE	FALSE	OPEN	20	0	19	12-Nov-25	C01	S12
193	FALSE	FALSE	OPEN	20	2	18	12-Nov-25	C01	S13
150	FALSE	FALSE	OPEN	20	19	1	12-Nov-25	C01	S14
30	FALSE	FALSE	OPEN	20	20	0	12-Nov-25	C01	S15
201	FALSE	FALSE	OPEN	20	12	6	12-Nov-25	C01	S16
187	FALSE	FALSE	OPEN	20	7	13	12-Nov-25	C01	S17
205	FALSE	FALSE	OPEN	20	10	8	12-Nov-25	C01	S18
72	FALSE	FALSE	OPEN	20	9	11	12-Nov-25	C01	S19

Data Modelling and Relationships: A Star Schema data model was created by connecting the fact table with the dimension tables using appropriate keys.



- One-to-many relationships between dimension and fact tables
- Fact table connected to all dimensions using keys
- Enables efficient filtering and analysis

8. **DAX Measures and Calculations:** To perform data analysis and generate meaningful insights, several DAX measures were created in Power BI.

```
Total no of Stations = COUNT(Dim_Station[Station-ID])
```

```
Total Availble Bike Stands = SUM(Fact_BikeStatus[Available Bike Stands])
```

```
Total Availble Bikes = SUM(Fact_BikeStatus[Available Bikes])
```

```
Total Bike Stands = SUM(Fact_BikeStatus[Bike Stands])
```

```
No of Closed Stations = CALCULATE(COUNT(Dim_Station[Station-ID]),FILTER
(Fact_BikeStatus,Fact_BikeStatus[Status]="CLOSED"))
```

```
No of Open Stations = CALCULATE(COUNT(Dim_Station[Station-ID]),FILTER
(Fact_BikeStatus,Fact_BikeStatus[Status]="OPEN"))
```

```
Station with Highest Bikes =
VAR T =
TOPN(
    1,
    SUMMARIZE(
        Dim_Station,
        Dim_Station[Station Name],
        "Bikes",[Total Availble Bikes]
    ),
    [Bikes], DESC
)
RETURN
MAXX(T, Dim_Station[Station Name])
```

```
Station with Lowest Bikes =
VAR T =
TOPN(
    1,
    SUMMARIZE(
        Dim_Station,
        Dim_Station[Station Name],
        "Bikes",[Total Availble Bikes]
    ),
    [Bikes], ASC
)
RETURN
MAXX(T, Dim_Station[Station Name])
```

```
No of Stations with banking facility = CALCULATE([Total no of Stations],FILTER(Fact_BikeStatus,
Fact_BikeStatus[Banking]="TRUE"))
```

```
No of Stations without banking facility = CALCULATE([Total no of Stations],FILTER(Fact_BikeStatus,
Fact_BikeStatus[Banking]="FALSE"))
```

```
No of Stations with bonus programs = CALCULATE([Total no of Stations],FILTER(Fact_BikeStatus,
Fact_BikeStatus[Bonus]="TRUE"))
```

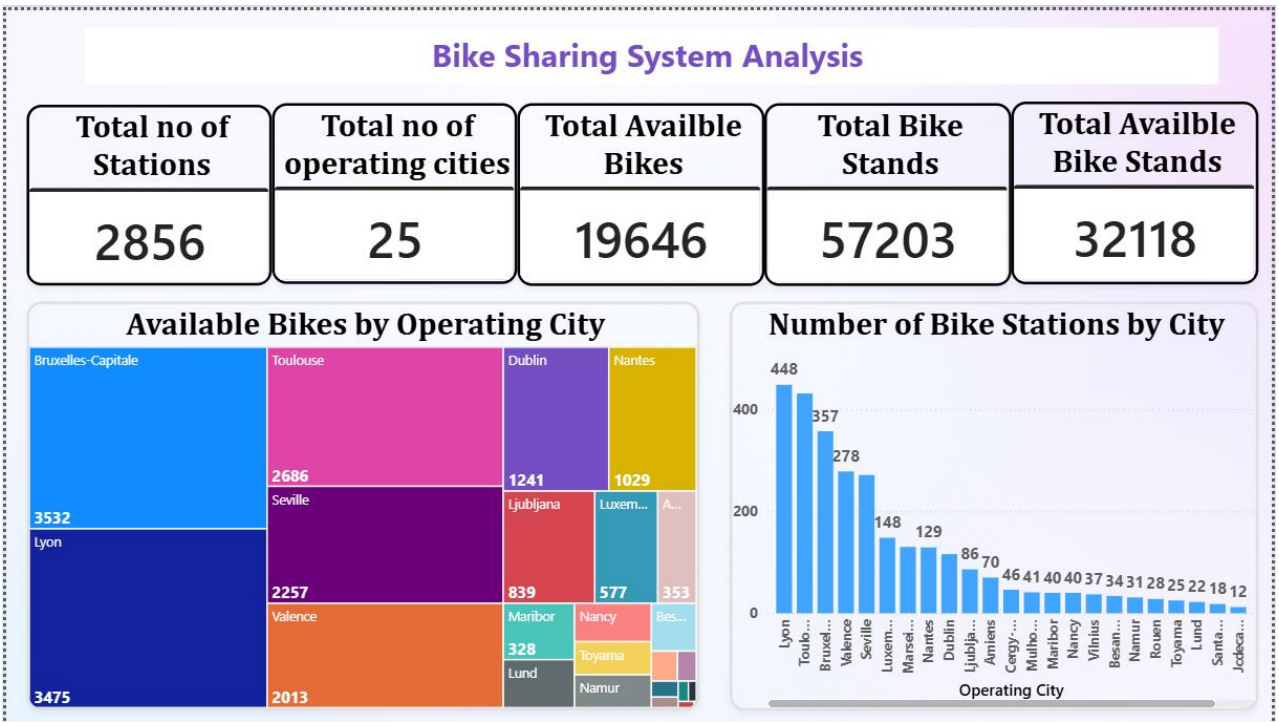
```
No of Stations without bonus programs = CALCULATE([Total no of Stations],FILTER(Fact_BikeStatus,
Fact_BikeStatus[Bonus]="FALSE"))
```

```
Total no of Operating Cities = DISTINCTCOUNT(Dim_City[City-ID])
```

Calculated column:

```
Operating City = RELATED(Dim_City[Operating City])
```

Dashboard and Visualization:

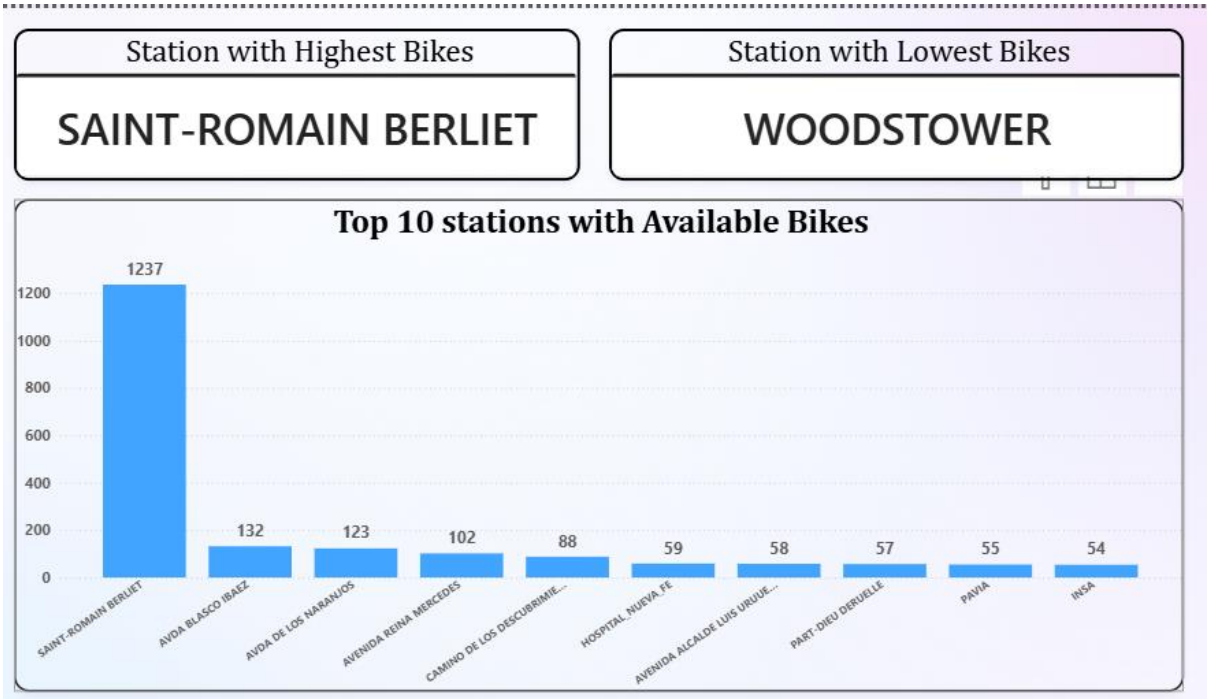


Insights:

- The system has 2856 stations across 25 cities.
- 19,646 bikes are available with 57,203 total stands.
- Lyon, Toulouse, and Bruxelles-Capitale have the most stations and bikes.
- Major cities have more stations due to higher demand and population.
- Smaller cities have **limited stations due to lower usage**.
- High-demand cities will likely need more bikes and stations in the future.
- Bike sharing may expand to smaller cities.
- Increase bikes in high-demand cities.
- Expand stations and improve bike redistribution for better availability.

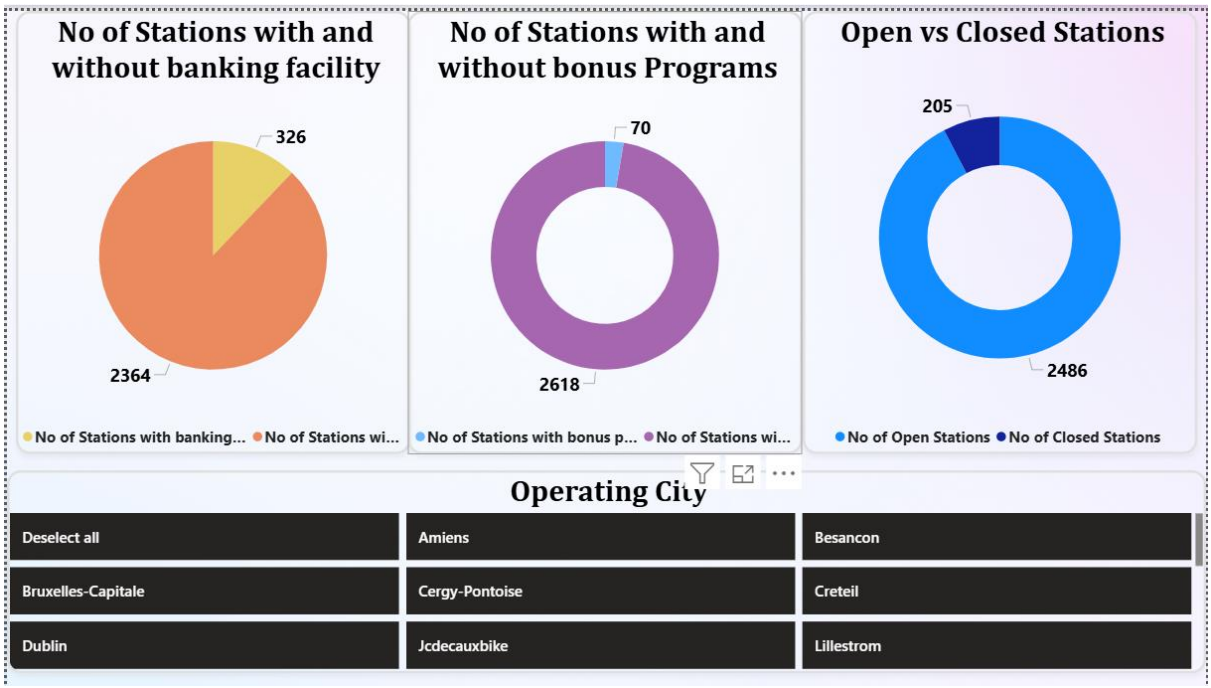
Insights:

- Bike stations are highly concentrated in Western Europe, especially France, Spain, Belgium, and Italy.
- The highest density of stations appears around France and nearby regions, indicating strong bike-sharing adoption.
- Few stations are present in Asia, with limited presence in Japan.
- The spatial distribution shows that the bike sharing system is mainly focused in European cities.



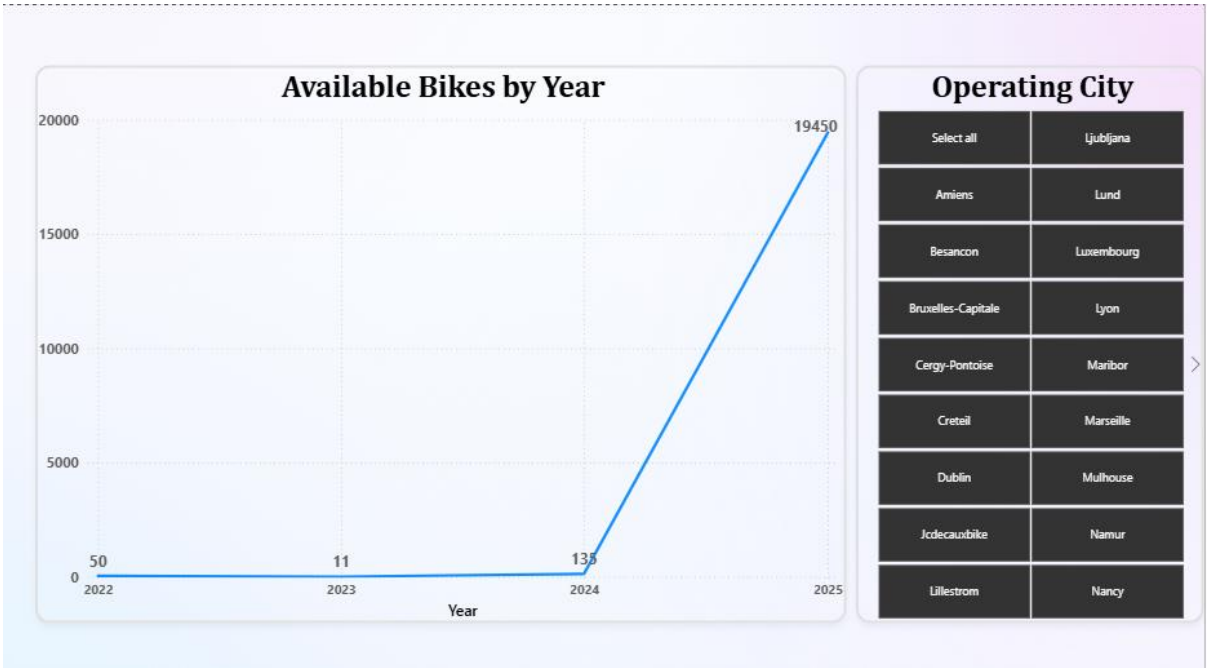
Insights:

- SAINT-ROMAIN BERLIET has the highest available bikes (1237).
- WOODSTOWER has the lowest bike availability.
- Other top stations have significantly fewer bikes compared to the highest station.
- The high bike count at SAINT-ROMAIN BERLIET may be due to larger station capacity or lower usage.
- Very low bikes at WOODSTOWER may indicate high demand or limited station capacity.
- Stations with very low bike availability may face shortages during peak hours.
- High-bike stations may continue to act as major supply points.
- Redistribute bikes from high-availability stations to low-availability stations.
- Monitor station demand to balance bike allocation efficiently.



Insights:

- 2364 stations do not have banking facility, while 326 stations have it.
- 2618 stations do not provide bonus facility, and only 70 stations provide it.
- 2486 stations are open, while 205 stations are closed.
- Most stations lack banking and bonus facilities due to limited infrastructure or operational policies.
- Closed stations may be due to maintenance, low demand, or operational issues.
- Stations without banking facilities may limit digital payment adoption in the future.
- Closed stations may reduce bike availability in certain areas if not reopened.
- Increase banking and bonus facilities to improve user convenience.
- Reopen or optimize closed stations to improve system coverage and accessibility.



City wise available bikes per each year

Operating City	2022	2023	2024	2025	Total
Amiens				353	**353**
Besancon			0	175	**175**
Bruxelles-Capitale			8	3524	**3532**
Cergy-Pontoise		0	4	13	**17**
Creteil		4	60		**64**
Dublin		1		1240	**1241**
Jcdecauxbike	1	2	3	17	**23**
Lillestrom		1		34	**35**
Ljubljana				839	**839**
Lund	3			276	**279**
Luxembourg		1		576	**577**
Lyon	0	0	2	3473	**3475**
Maribor	0			328	**328**
Marseille	46				**46**
Mulhouse			0	13	**13**
Namur		0	2	204	**206**
Nancy	0			241	**241**
Nantes	0		0	1029	**1029**
Total	**50**	**11**	**135**	**19450**	**19646**

Station wise available bikes per each year

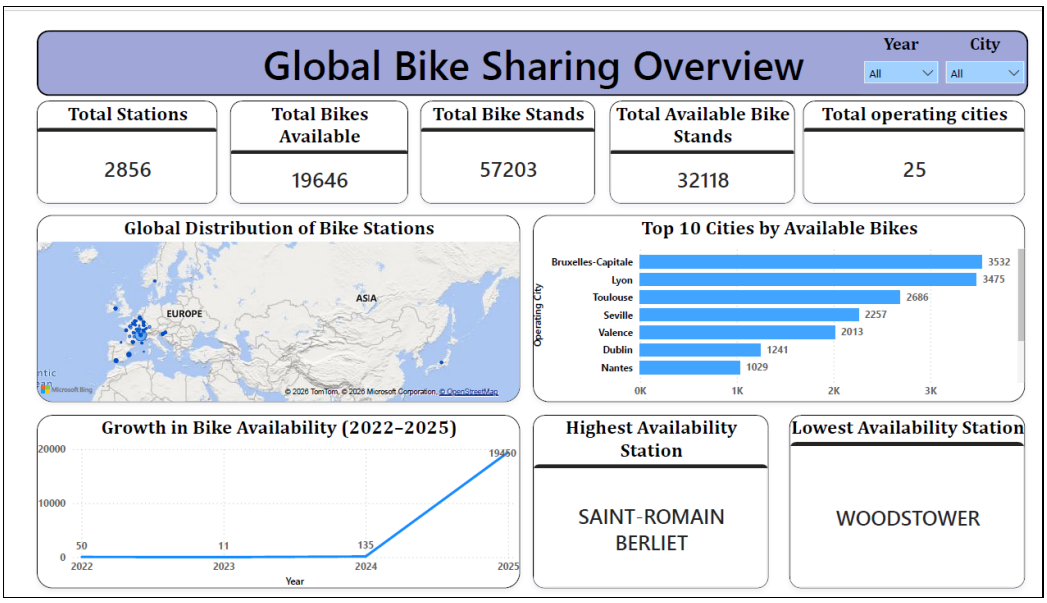
Station Name	2022	2023	2024	2025	Total
SAINT-ROMAIN BERLIET	4	4	2	1227	**1237**
AVDA BLASCO IBAEZ				132	**132**
AVDA DE LOS NARANJOS				123	**123**
AVENIDA REINA MERCEDES				102	**102**
CAMINO DE LOS DESCUBRIMIENTOS				88	**88**
HOSPITAL_NUEVA_FE				59	**59**
AVENIDA ALCALDE LUIS URUUELA				58	**58**
PART-DIEU DERUELLE				57	**57**
PAVIA				55	**55**
INSA				54	**54**
Total	**50**	**11**	**135**	**19450**	**19646**

Insights:

- Available bikes increased from 50 (2022) to 19,450 (2025).
- The highest bike availability is in 2025.
- The sharp increase due to stations like **Lyon and Bruxelles-Capitale** show higher bike availability.
- If the trend continues, bike availability will keep increasing in future years.
- Authorities should expand stations and manage bike distribution to support the growing system.

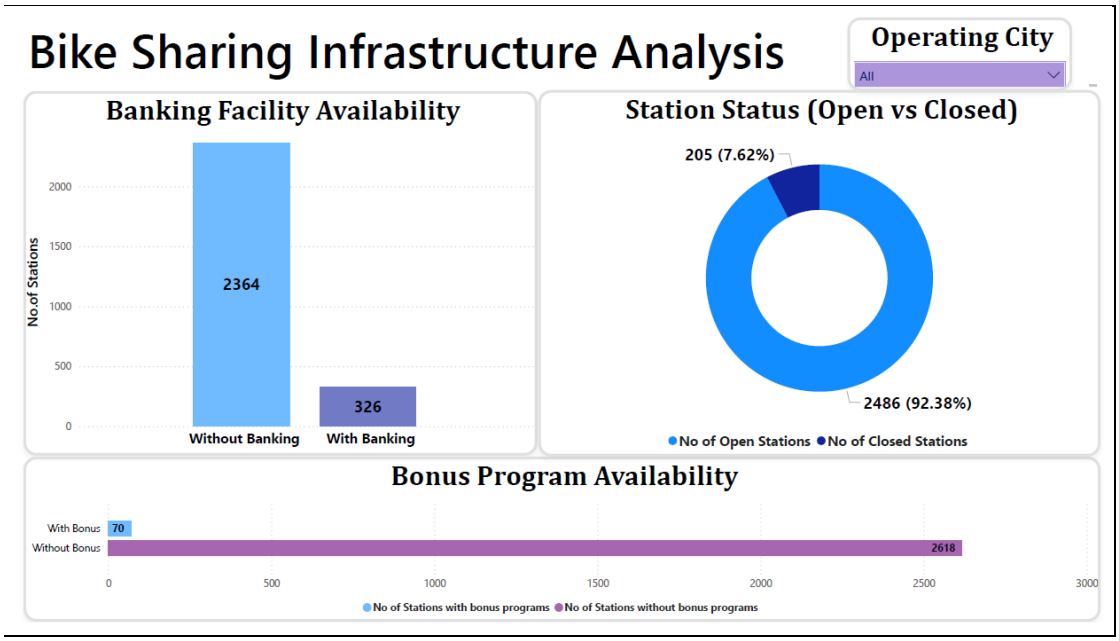
Dashboards

Overview Dashboard:



- This dashboard provides a high-level overview of the bike-sharing system across multiple cities. It highlights key metrics such as total stations, bike availability, and operating cities.
- The dashboard also visualizes geographic distribution, top-performing cities, growth trends over time, and identifies stations with highest and lowest availability.

Infrastructure Dashboard:



- This dashboard focuses on operational insights by analyzing infrastructure availability across stations.
- It highlights the presence of banking facilities, bonus programs, and station status (open vs closed). These insights help identify infrastructure gaps and areas for improvement in the bike-sharing system.

9. Final Conclusion:

- This project was analyzed using **Power BI**, where bike-sharing data was transformed into meaningful insights through data cleaning, modelling, and visualization.
- The analysis revealed an imbalance in bike availability across stations, with higher demand in major cities, limited digital infrastructure, and underutilized bonus systems.
- To improve efficiency, strategies such as smart bike redistribution, demand-based expansion, digital payment integration, and incentive-driven (bonus) programs can be implemented to enhance availability and user experience